

MATH 554: Linear Algebra Qualifying Exam Prep

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Part I

Notes

These notes are based on the MA 554: Linear Algebra class taught by Dr. Jeremy Miller in Fall 2025 at Purdue. The first half of the notes are based on [1], and the rest are based on [2], all united through Dr. Miller's notes in [3]. They are meant to prepare for the qualifying exam in Linear algebra. Unless specified otherwise, suppose R is a ring, I, J are ideals, V is a vector space, and M is a module.

1 Preliminary Definitions

Definition 1.1: Group.

A group is a set G and maps $G \times G \rightarrow G$ such that

- (a) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in G$.
- (b) There exists an element e with $a = ae = ea$ for all $a \in G$.
- (c) For any $a \in G$, there exists an element $a^{-1} \in G$ such that $aa^{-1} = a^{-1}a = e$.

A group is considered abelian if it is also commutative. Groups are sets with a single operation, but we can extend this to two operations by considering rings.

Definition 1.2: Ring.

A ring R is a set with two operations, $+, \times : R \times R \rightarrow R$ with

- (a) $(R, +)$ is an abelian group with $e = 0$.
- (b) $a \times (b \times c) = (a \times b) \times c$ for all $a, b, c \in R$.
- (c) There exists $1 \in R$ with $1 \times a = a \times 1 = a$ for all a .
- (d) $(a + b) \times c = a \times c + b \times c$ and $c \times (a + b) = c \times a + c \times b$ for all $a, b, c \in R$.

Rings are considered commutative if their $\times$ operation is commutative. We can further define sets with multiple operations by considering Fields.

Definition 1.3: Field.

A field is a set F and two operations $+, \cdot : F \times F \rightarrow F$ such that

- (a) $a + (b + c) = (a + b) + c$ for all $a, b, c \in F$
- (b) $a + b = b + a$ for all $a, b \in F$
- (c) There exists $0, 1 \in F$ with $0 + a = a$ and $1 \cdot a = a$ for all $a \in F$.
- (d) $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ for all $a, b, c \in F$.
- (e) $ab = ba$ for all $a, b \in F$.
- (f) For all $a \neq 0$, there exists an a^{-1} such that $aa^{-1} = a^{-1}a = 1$.

Definition 1.4: Vector Space.

Let $\mathbb{F}$ be a field. An $\mathbb{F}$ -vector space is a set V and two operations $+: V \times V \rightarrow V$ and $\cdot : \mathbb{F} \times V \rightarrow V$ such that

- (a) $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$ for all $\vec{a}, \vec{b}, \vec{c} \in V$.
- (b) There exists a $\vec{0} \in V$ with $\vec{0} + \vec{a} = \vec{a}$.
- (c) For all $\vec{a} \in V$, there exists $-\vec{a} \in V$ with $\vec{a} + (-\vec{a}) = \vec{0}$.
- (d) $r(\vec{a} + \vec{b}) = r\vec{a} + r\vec{b}$ for all $\vec{a}, \vec{b} \in V$ and $r \in \mathbb{F}$.
- (e) $(r + s)\vec{a} = r\vec{a} + s\vec{a}$ for all $r, s \in \mathbb{F}$ and $\vec{a} \in V$.

Note that a ring as in Definition 1.2 is a set satisfying all axioms of a field in Definition 1.3 except for (e) and (f). If it is a commutative ring, then it satisfies (e).

Example 1.1

$\mathbb{Z}$ is a commutative ring, but not a field.

Definition 1.5: Left Module.

Let R be a ring. A left R -module is a set with two operations satisfying the conditions of Definition 1.4

Definition 1.6: Ideal.

Let R be a commutative ring. A set $I \subseteq R$ is an ideal if

- (a) $0 \in I$.
- (b) For all $a, r \in I$, we have $ar \in I$.
- (c) For all $a, b \in I$, then $a + b \in I$.

Definition 1.7: Subring.

Let R be a ring. A set $K \subseteq R$ is a subring if

- (a) $0, 1 \in K$.
- (b) For all $a, r \in K$, we have $ar \in K$.
- (c) For all $a, b \in K$, we have $a + b \in K$.

Proposition 1.8.

A commutative ring R is a field if and only if $\{0\}, R$ are the only ideals.

Definition 1.9.

Let R be a ring. Then define

$$R[x] = \{r_1 + r_1x + r_2x^2 + \cdots + r_nx^n : r_0, \dots, r_n \in R\} \quad (1.1)$$

Proposition 1.10.

If R is commutative, then so is $R[x]$.

Theorem 1.11.

If $\mathbb{F}$ is a field, all ideals in $\mathbb{F}[x]$ are of the form $(f(x)), f(x) \in \mathbb{F}[x]$.

Lemma 1.12.

Let $f(x), g(x) \in \mathbb{F}[x]$. Then there exist some $d(x), r(x)$ with $\deg r(x) < \deg f(x)$ such that

$$g(x) = d(x)f(x) + r(x) \quad (1.2)$$

Definition 1.13.

Let V, W be $\mathbb{F}$ -vector spaces. Then $f : V \rightarrow W$ is linear if

- (a) $f(\vec{x} + \vec{y}) = f(\vec{x}) + f(\vec{y})$ for all $\vec{x}, \vec{y} \in V$.
- (b) $f(r\vec{x}) = rf(\vec{x})$ for all $\vec{x} \in V$ and $r \in \mathbb{F}$.

Proposition 1.14.

The data of an $\mathbb{F}[x]$ -module is a vector space V and a linear map $f : V \rightarrow V$.

Definition 1.15: Monoid.

A monoid is a set M and a map $\cdot : M \times M \rightarrow M$ such that

- (a) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in M$.
- (b) There exists a $1 \in M$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in M$.

Note that a monoid is a group as seen in Definition 1.1, except it does not necessarily include inverses.

Example 1.2

$(\mathbb{N}, +)$ is a monoid, but not a group. It clearly has associativity and the identity since $1 \in \mathbb{N}$, but there are no inverses since we would need -1 , but $1 \notin \mathbb{N}$ so it is a monoid but not a group.

Definition 1.16.

Let R be a ring, and M a monoid. Define

$$R[M] = \{r_1m_1 + r_2m_2 + \cdots + r_nm_n : r_i \in R, m_i \in M\} \quad (1.3)$$

Definition 1.17.

Let M, N be monoids. $f : M \rightarrow N$ is a map with

$$f(m_1m_2) = f(m_1)f(m_2) \quad (1.4)$$

for all $m_1, m_2 \in M$.

Proposition 1.18.

Let V be an R -module. A $R[M]$ -module structure on V is a monoid homomorphism $M \rightarrow \text{hom}_R(V, V)$. If M is a group, then $M \rightarrow \text{Aut}_R(V, V)$.

Proposition 1.19.

$\text{hom}_R(V, V)$ is a ring with $f : V \rightarrow V$ and $g : V \rightarrow V$ such that

$$(f + g)(v) = f(v) + g(v) \quad (1.5)$$

for all $v \in V$.

Definition 1.20: Ring Homomorphism.

Let R, R' be rings. A function $f : R \rightarrow R'$ is a ring homomorphism if

$$(a) \ f(r_1r_2) = f(r_1)f(r_2) \text{ for all } r_1, r_2 \in R.$$

$$(b) \ f(r_1 + r_2) = f(r_1) + f(r_2) \text{ for all } r_1, r_2 \in R.$$

Definition 1.21: Isomorphism.

A homomorphism $f : R \rightarrow R'$ is an isomorphism if there is an inverse homomorphism $f^{-1} : R' \rightarrow R$ such that

$$f^{-1} \circ f(x) = x \quad (1.6)$$

for all $x \in R$. If it exists, we say $R \cong R'$.

Theorem 1.22.

let $\mathbb{F}$ be a field. Then $\text{Mat}_{\mathbb{F}}(n, n) \cong \text{hom}_{\mathbb{F}}(\mathbb{F}^n, \mathbb{F}^n)$.

Definition 1.23.

Let $f : R \rightarrow R'$ be a ring homomorphism. Then define the kernel of f as

$$\ker(f) = \{r \in R : f(r) = 0\} \quad (1.7)$$

Proposition 1.24.

$\ker(f) \subseteq R$ is an ideal.

Definition 1.25.

Let $f : R \rightarrow R'$ be a ring homomorphism. Then

$$\text{im}(f) = \{x | f(r) = x, \forall r \in R\} \quad (1.8)$$

Proposition 1.26.

$\text{im}(f) \subseteq R'$ is a subring.

Theorem 1.27: First Isomorphism Theorem.

Let $f : R \rightarrow S$ be a ring homomorphism. Then

$$R/\ker(f) \cong \text{im}(f) \quad (1.9)$$

Proposition 1.28.

If R is a ring, then there exists a unique $f : \mathbb{Z} \rightarrow R$ which is a ring homomorphism.

Definition 1.29.

Let A be an abelian group and $B \subseteq A$ a subgroup. Define the equivalence class of $a \in A$ be

$$[a] = \{a + b : b \in B\} \quad (1.10)$$

We define a quotient group by

$$A/B = \{[a] : a \in A\} \quad (1.11)$$

Proposition 1.30.

Let $I \subseteq R$ be an ideal. Then the map $f : R \rightarrow R/I$ is surjective.

Proposition 1.31.

Let $K \subseteq R$ be a subring. Then the map $g : K \rightarrow R$ is injective.

2 Domains and Fields of Fractions

Definition 2.1: Maximal Ideal.

$I \subset R$ is called a maximal ideal if

1. $I \neq R$
2. $J \subseteq I$ implies that either $J = R$ or $J = I$.

Proposition 2.2.

Let $\mathbb{F}$ be a field. $\mathbb{F}$ and $\{0\}$ are the only ideals of $\mathbb{F}$.

Corollary 2.3.

0 is a maximal ideal in $\mathbb{F}$.

Definition 2.4: Integral Domain.

R is an integral domain if

- (a) R is commutative
- (b) If $ab = 0$ then $a = 0$ or $b = 0$.

Corollary 2.5.

If R is commutative and $M \subsetneq R$ is maximal, then M is prime.

Definition 2.6: Field of Fractions.

Let R be a commutative ring and $S \subseteq R$ a multiplicatively closed subset. Assume $1 \in S$. Let $S^{-1}R = (R \times S)/\sim$ defining an equivalence relation by saying

$$\frac{r}{s} \sim \frac{r'}{s'} \quad (2.1)$$

if there exists a nonzero t such that $trs' = tr's$. This t can be ignored if R is an integral domain.

Example 2.1

If $S = \{0, 1\}$ then $S^{-1}R = 0$ and $\frac{r}{s} = \frac{0}{0}$.

Example 2.2

If $R = \mathbb{Z}$ and $S = \mathbb{Z} - \{0\}$ then $S^{-1}R = \mathbb{Q}$.

Definition 2.7.

Define $\mathbb{F}(x) = S^{-1}\mathbb{F}[x] = \left\{ \frac{f(x)}{g(x)} : g(x) \neq 0 \right\}$ with $S = \mathbb{F}[x] - \{0\}$.

Theorem 2.8.

If $f(x) \in \mathbb{F}[x, x^{-1}]$ has an inverse, then $f(x) = ax^k$ for $a \neq 0$ and $k \in \mathbb{Z}$.

Corollary 2.9.

$$\frac{1}{x+1} \notin \mathbb{F}[x, x^{-1}]$$

Definition 2.10.

For $s \in \{1, a, a^2, \dots\} \subset R$, write $s^{-1}R = R[a^{-1}]$.

Example 2.3

$$\mathbb{Z}[2^{-1}] = \left\{ \frac{x}{2^n} \right\} \subseteq \mathbb{Q}$$

Example 2.4

$$\mathbb{Z}[2^{-1}][3^{-1}] \cong \mathbb{Z}[6^{-1}]$$

Proposition 2.11.

$S^{-1}R$ is a ring.

Definition 2.12.

Let $\iota : R \rightarrow S^{-1}R$ be $\iota(t) = \frac{t}{1}$.

Proposition 2.13.

$\iota : R \rightarrow S^{-1}R$ is a ring homomorphism.

Proposition 2.14.

If R is an integral domain, then ι is injective.

Proof. Suppose $\iota(r) = 0$. Then $\frac{r}{1} = \frac{0}{1}$ meaning that $t \cdot r \cdot 1 = t \cdot 1 \cdot 0$ for some $t \neq 0$. This implies that $r = 0$ or $t = 0$, but since t cannot be zero, so ι must be injective. $\square$

Proposition 2.15: Universal Mapping Property.

Suppose $S \subset R$ and let $f : R \rightarrow R'$ be a ring homomorphism. Suppose for all $x \in S$, $f(x)$ has an inverse. Then there exists a unique $g : S^{-1}R \rightarrow R'$ with $g \circ \iota = f$.

Definition 2.16.

If R is an integral domain, let $\text{Frac}(R) = S^{-1}R$ with $S = R - \{0\}$.

Example 2.5

$$\text{Frac}(\mathbb{F}[x]) = \mathbb{F}(x)$$

Example 2.6

$$\text{Frac}(\mathbb{Z}) = \mathbb{Q}$$

Proposition 2.17.

$\text{Frac}(R)$ is a field. Proposition 2.15 shows that this is the smallest field that you can embed your ring into.

Corollary 2.18.

If R is an integral domain, $\mathbb{F}$ is a field, and $f : R \rightarrow \mathbb{F}$ is injective, then there exists a function $g : \text{Frac}(R) \rightarrow \mathbb{F}$ injective with

Proposition 2.19.

Let $\mathbb{F}, \mathbb{K}$ be fields and $g : \mathbb{F} \rightarrow \mathbb{K}$ be a homomorphism. Then g is injective.

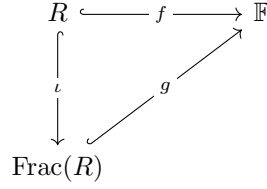


Figure 1

Proof. Suppose $g(a) = 0$. Then

$$1_{\mathbb{K}} = g(1_{\mathbb{F}}) = g(aa^{-1}) = 0g(a^{-1}) = 0 \quad (2.2)$$

which is a contradiction, so $a = 0$ and g is injective. $\square$

Definition 2.20.

If M is an R -module, $1 \in S \subseteq R$, S is closed under multiplication, and R is commutative, then let

$$S^{-1}M = M \times S / \sim \quad (2.3)$$

with $\frac{m}{s} \sim \frac{m'}{s'}$ if $tms' = tm's$ for $t \neq 0$.

Proposition 2.21.

$S^{-1}M$ is a $S^{-1}R$ -module.

2.1 Integral Domains and Decompositions

Definition 2.22: Euclidean Domain.

Let R be an integral domain. We call R Euclidean if there is $N : R \rightarrow \mathbb{N}$ such that for all $r \in R$

(a) $N(r) = 0$ if and only if $r = 0$

(b) for all $a, b \in R$ with $b \neq 0$ there exists some d, r with $a = db + r$ and $N(r) < N(b)$.

Theorem 2.23.

If $\mathbb{F}$ is a field, then $\mathbb{F}[x]$ is Euclidean with $N = \deg + 1$.

Theorem 2.24.

If $\mathbb{F}$ is a field, then $\mathbb{F}$ is Euclidean.

Definition 2.25: Principle Ideal Domain.

Let R be an integral domain. R is a PID if all ideals are of the form (r) for $r \in R$.

Theorem 2.26.

If R is Euclidean, then R is a PID.

Proof. Let $I \subset R$ be an ideal. Assume $I \neq 0$. Let $D = \min_{i \in I, i \neq 0} N(i) \in \mathbb{N}_0$ and $D > 0$. Pick b with $N(b) = D$. Consider $a \in I$. We want to show that $a \in (b)$, so then $I = (b)$. We can write $a = db + r$ with $N(r) < N(b)$ for some $r \in I$, so then

$$0 = N(r) < N(b) = D \quad (2.4)$$

Which implies that $r = 0$ so $a = db$ meaning $a \in (b)$. $\square$

Corollary 2.27.

$\mathbb{Z}[x]$ is not Euclidean.

Definition 2.28.

u is a unit if there exists some w such that

$$uw = wu = 1 \quad (2.5)$$

Definition 2.29.

Let R be commutative, and $a, b \in R$ with $b \neq 0$. We write $a|b$ if there exists some $c \in R$ with $ac = b$.

Definition 2.30.

Let R be a commutative ring. We say that x is the $\gcd$ of $a, b \in R$ if

$$(a) \ x|a$$

$$(b) \ x|b$$

$$(c) \text{ for all } y, y|a \text{ and } y|b \text{ implies } y|x$$

Definition 2.31.

For $a_1, \dots, a_k \in R$, let

$$(a_1, \dots, a_k) = \{r_1 a_1 + \dots r_k a_k : r_i \in R\} \quad (2.6)$$

Proposition 2.32.

If $(a, b) = (x)$ then $x = \gcd(a, b)$.

Proof. First, assume $y|a$ and $y|b$. We want to show that $y|x$. We know that $a \in (y)$ and $b \in (y)$ so $x = (a, b) \subset (y)$. So then $x \in (y)$ so $y|x$. $\square$

Proposition 2.33.

If R is an integral domain and $(d) = (d')$, then $d' = du$ for a unit u .

Proof. Assume $d, d' \neq 0$. we have that $d \in (d) = (d')$ so $d = xd'$ and $d' = yd$ for some $x, y \in R$. So then

$$d = xyd \quad (2.7)$$

$$0 = (1 - xy)d \quad (2.8)$$

Since $d \neq 0$, $1 - xy = 0$ meaning that $xy = 1$ so x and y are units. $\square$

Corollary 2.34.

If R is a PID, then $\gcd$ exists and are unique up to a unit.

In a Euclidean domain, it is of interest to directly compute $\gcd(a, b)$. To this end, we have a scheme:

1. Assume $N(a) \leq N(b)$ and $r_0 = b$ and $r_1 = a$.
2. Write $r_i = dr_{i+1} + r_{i+2}$ with $N(r_{i+2}) \leq N(r_{i+1})$.
3. Repeat until you find a k such that $r_k = 0$.
4. Then $r_{k-1} = \gcd(a, b)$.

This is essentially applying the Euclidean Algorithm repeatedly until you find the smallest generator of the elements.

Definition 2.35: Ideal Addition.

For $I, J \subseteq R$ be ideals. Define

$$I + J = \{i + j : i \in I, j \in J\} \quad (2.9)$$

Proposition 2.36.

For I, J ideals, $I + J$ is also an ideal.

Proposition 2.37.

If R is a PID with ideals $(a), (b)$, then

$$(a) + (b) = (\gcd(a, b)) \quad (2.10)$$

Definition 2.38.

Let R, S be rings. $R \times S$ is a ring with

$$(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2) \quad (2.11)$$

$$(r_1, s_1)(r_2, s_2) = (r_1 r_2, s_1 s_2) \quad (2.12)$$

Note that $R \times S$ will not necessarily be an integral domain if $1 \neq 0$ in R and S .

Definition 2.39.

I and J are comaximal if $I + J = R$.

Example 2.7

$(6) + (25) = \mathbb{Z}$, so $(6), (25)$ are comaximal ideals in $R = \mathbb{Z}$.

Definition 2.40: Ideal Multiplication.

Let R be a commutative ring and I, J ideals. Define

$$IJ = \left\{ \sum i_n j_n : i_n \in I, j_n \in J \right\} \quad (2.13)$$

Proposition 2.41.

IJ is an ideal.

Proposition 2.42.

$IJ \subseteq I \cap J$

Proof. Let $x \in IJ$. We want to show that $x \in I$ and $x \in J$. We know that $x = \sum i_n j_n$, and we want to show that each term $i_n j_n$ in the sum is in both I and J . $\square$

Theorem 2.43: Chinese Remainder Theorem.

Let $A_1, \dots, A_k \subseteq R$ be ideals. Let $f : R \rightarrow R/A_1 \times \dots \times R/A_k$ be defined as

$$f(r) = ([r], \dots, [r]) \quad (2.14)$$

If $A_i + A_j = R$ for all $i \neq j$, then

$$(a) \ker(f) = \bigcap_{i=1}^k A_i$$

(b) f is surjective

$$(c) \bigcap_{i=1}^k A_i = A_1 A_2 \dots A_k$$

Proof. (a) An element $r \in \ker(f)$ if and only if $f(r) = 0$ which can happen if and only if $([r], \dots, [r]) = ([0], \dots, [0])$ which is true if and only if $r \in A_1 \cap A_2 \cap \dots \cap A_k$ which is what we want.

(b) For $k = 2$, assume $A + B = R$. We want to show that $f : R \rightarrow R/A \times R/B$ is surjective. Pick some $[r] \in R/A$ and $[s] \in R/B$. Then $A + B = R$ implies that there are some $a \in A$ and $b \in B$ with $a + b = 1$. So then we see

$$f(ra + sb) = ([ra + sb], [ra + sb]) \quad (2.15)$$

We need to check if this is in $[r]$, so we take $ra + sb - r$ and see

$$[ra + sb]_A = [sb] = [s(1 - a)] = [s - as] = [s] \in A \quad (2.16)$$

so we are finished. We can proceed by induction on k , which is left as an exercise.

(c) Again, assume $k = 2$ so we need to show that $A \cap B = AB$. Clearly $AB \subseteq A \cap B$, so we only need to show that $A \cap B \subseteq AB$. Pick some $c \in AB$. We then see

$$c = 1c = (a + b)c = ac + bc \in AB \quad (2.17)$$

which proves the $k = 2$ case. We can again induct on k to get the rest. $\square$

Example 2.8

Let $R = \mathbb{Z}$, $A_1 = (2)$, $A_2 = (3)$. Then $f : \mathbb{Z} \rightarrow \mathbb{Z}/2 \times \mathbb{Z}/3$ gives us

$$A_1 + A_2 = (2) + (3) = (1) = \mathbb{Z} \quad (2.18)$$

$$A_1 A_2 = (2)(3) = (6) \quad (2.19)$$

and

$$\ker(f) = A_1 A_2 = (6) \quad (2.20)$$

$$\operatorname{im}(f) = \mathbb{Z}/2 \times \mathbb{Z}/3 \quad (2.21)$$

So then by Theorem 1.27, we have that $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$.

Example 2.9

$$\mathbb{R}[x]/(x^2 + 1) = \mathbb{C}$$

3 Modules and Tensor Products

We will now consider modules and their applications to linear algebra.

Definition 3.1: Addition of Submodules.

Let $A_1, \dots, A_k \subseteq M$ be submodule. Then we define

$$A_1 + \dots + A_k = \{a_1 + \dots + a_k : a_i \in A_i\} \quad (3.1)$$

Proposition 3.2.

$A_1 + \dots + A_k$ is a submodule.

Example 3.1

If $R = \mathbb{Z}$ and $M = \mathbb{Z}^2$ and

$$A = \left\{ \begin{bmatrix} x \\ 0 \end{bmatrix} : 2|x \right\} = \operatorname{Span} \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad (3.2)$$

$$B = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x = y \right\} = \operatorname{Span} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (3.3)$$

Then $A + B \cong \mathbb{Z}^2$ and $M/A + B \cong \mathbb{Z}/2$.

Theorem 3.3.

Let M and N be R -modules and $A, B \subseteq M$ be submodules. Then

- (a) For $f : M \rightarrow N$ which is R linear, then $M/\ker f \cong \operatorname{im} f$.
- (b) $(A + B)/B \cong A/A \cap B$
- (c) If $A \subseteq B$, then $(M/B)/(B/A) \cong M/B$.
- (d) $\{M \supseteq C \supseteq A\} \cong \{M/A \supseteq C' \supseteq 0\}$ for some C, C' submodules where $\pi : M \rightarrow M/A$ defines $C \mapsto C/A$ and $C' \mapsto \pi^{-1}(C')$.

Definition 3.4: Span of a Module.

Let $S \subseteq M$ be a set and M an R -module. Then

$$\operatorname{Span}_R(S) = \langle S \rangle_R = \left\{ \sum r_i s_i : r_i \in R, s_i \in S \right\} \quad (3.4)$$

We say that S generates M if $\langle S \rangle = M$. M is finitely generated if $M = \langle s_1, \dots, s_n \rangle$ for some finite n . We say that M is cyclic if $M = \langle S \rangle$ for some S .

Definition 3.5: External Direct Sum.

Let M and N be R -Modules. Let $M \oplus N = M \times N$, and define

$$(n_1, n_1) + (m_2, n_2) = (n_1 + m_1, n_2 + m_2) \quad (3.5)$$

$$r(n_1, n_1) = (rn_1, rn_2) \quad (3.6)$$

Definition 3.6: Internal Direct Sum.

Let $A, B \subseteq M$ be submodules. We say that $M = A \oplus_{\text{internal}} B$ if there is some map $A \oplus B \rightarrow M$ such that $(a, b) \mapsto a + b$ is an isomorphism.

Definition 3.7: Free Modules.

For a set S , a ring R , we define the free R -module over S , $F_R(S)$ as

$$F_R(S) = \{f : S \rightarrow R \mid f(s) = 0 \text{ for all but finitely many } s \in S\} \quad (3.7)$$

Definition 3.8.

There is a natural map $\iota : S \rightarrow F_R(S)$ such that for all $s \in S$ we have $\iota(s) : S \rightarrow R$. We can define

$$(\iota(s))(t) = \begin{cases} 1 & s = t \\ 0 & s \neq t \end{cases} \quad (3.8)$$

This acts as a sort of "indicator" function. Using this, we can establish a universal mapping property for free modules.

Theorem 3.9: Universal Mapping Property for Free Modules.

Let S be a set, M be an R -module and $f : S \rightarrow M$, then there exists a unique $\hat{f} : F_R(S) \rightarrow M$ which is R -linear and makes Figure 2 commute.

$$\begin{array}{ccc} S & \xrightarrow{f} & M \\ \downarrow \iota & \nearrow \hat{f} & \\ F_R(S) & & \end{array}$$

Figure 2

This process is analogous to how we can characterize a linear operator by its action on a basis, but now we are associating a set with a free module rather than a basis in the traditional sense. We can now define a basis for an R -module.

Definition 3.10.

Let M be an R -module. We say that a set $S \subseteq M$ is a basis if for any map i we have that Figure 3 implies $F_R(S) \cong M$.

$$\begin{array}{ccc} S & \xrightarrow{i} & M \\ \downarrow \iota & \nearrow \hat{i} & \\ F_R(S) & & \end{array}$$

Figure 3

Example 3.2

If we let $R = \mathbb{R}$ and $M = \mathbb{R}^2$ we can let

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

We have that $F_{\mathbb{R}}(S) \rightarrow \mathbb{R}^2$ is an isomorphism, meaning S is a basis for $\mathbb{R}^2$, which is consistent to what we know from the theory of basis' in linear algebra.

3.1 Tensor Products

Definition 3.11: Tensor Product.

Let R be a ring, M a right R -module and N a left R -module. Then we define

$$M \otimes_R N = F_{\mathbb{Z}}(M \times N)/A \quad (3.9)$$

where

$$A = \langle (mr) \times n - m \times (rn), (m_1 + m_2) \times n - (m_1 \times n) - (m_2 \times n), m \times (n_1 + n_2) - m \times n_1 - m \times n_2 \rangle \quad (3.10)$$

for some $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$ and $r \in R$. We write $m \otimes n$ for the image of $m \times n \in M \times N$ in $F_{\mathbb{Z}}(M \times N)$. We also write for $x \in M \otimes N$

$$x = \sum_i k_i (m_i \otimes n_i), \quad k_i \in \mathbb{Z}, m_i \in M, n_i \in N \quad (3.11)$$

Finally, we have the following rules for all elements $m \in M$, $n \in N$ and $r \in R$:

$$(mr) \otimes n = m \otimes (rn) \quad (3.12)$$

$$(m_1 + m_2) \otimes n = (m_1 \otimes n) + (m_2 \otimes n) \quad (3.13)$$

$$m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2 \quad (3.14)$$

Example 3.3

We want to show that

$$\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}_3 = 0$$

We can repeatedly apply (3.12) to $1 \otimes 1$ to see

$$1 \otimes 1 = 1 \cdot 3 \otimes 1 = 1 \otimes 0 = 0 \otimes 0 = 0$$

Proposition 3.12.

$M \otimes_R N$ is an abelian group.

Definition 3.13: Balanced Map.

Let M be a right R -module, N a left R -module and B an abelian group. A function $f : M \times N \rightarrow B$ is called R -balanced if :

- (a) $f(\cdot, n) : M \rightarrow B$ is $\mathbb{Z}$ -linear for all $n \in N$.
- (b) $f(m, \cdot) : N \rightarrow B$ is $\mathbb{Z}$ -linear for all $m \in M$.
- (c) $f(mr, n) = f(m, rn)$ for all $m \in M, n \in N, r \in R$.

Note that balanced maps are not necessarily homomorphism, but are functions in a groups of one variable.

Theorem 3.14: Universal Mapping Property for Balanced Maps.

Let B be an abelian group, and $\iota : M \times N \rightarrow N \otimes N$ be $\iota(m \times n) = m \otimes n$.

Theorem 3.15: Universal Mapping Property for Quotients.

Let B be an abelian group and $A \subset B$ be a subgroup. Let $\pi : B \rightarrow B/A$ be $\pi(b) = [b]$. Let A, B, C be abelian groups with $A \subset B$ a subgroup. Let $f : B \rightarrow C$ be a map of abelian groups with $f(a) = 0$ for all $a \in A$. Then there exists a unique map of abelian groups $\hat{f} : B/A \rightarrow C$ such that $\hat{f}(\pi())$

Definition 3.16: Bimodule.

Let R, S be rings and M a left R -module and a right S -module. We say that M is a $R - S$ -bimodule if

$$(rm)s = r(ms) \quad (3.15)$$

for all $r \in R$, $m \in M$, and $s \in S$.

Note that if M is a left R -module for a commutative ring R , then M is an R -bimodule with $mr = rm$ for all $r \in R$ and $m \in M$.

Example 3.4

Let J is a two sided ideal in R . Then J is an R -bimodule.

Definition 3.17.

Let M be an $R - S$ bimodule and N a left S -module. Then $M \otimes_S N$ is a left R -module via

$$r(m \otimes n) = (rm) \otimes n \quad (3.16)$$

Corollary 3.18.

If R is commutative, then $M \otimes_R N$ is an R -module.

Example 3.5

Let $R = \mathbb{R}[x]$, $M = \mathbb{R}^2$, and $N = \mathbb{R}$. Let x act on $\mathbb{R}^2$ via $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and x act on $\mathbb{R}$ via $[0]$. We want to find the structure of $\mathbb{R} \otimes_{\mathbb{R}[x]} \mathbb{R}^2$. To do this, we can calculate

$$e_1 \otimes 1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} e_1 \otimes 1 = e_1 x \otimes 1 = e_1 \otimes x1 = e_1 \otimes 0 = 0$$

$$e_2 \otimes 1 = ???$$

Supposedly this generalizes to Proposition 3.19 and we get $\mathbb{R}^2 \otimes_{\mathbb{R}[x]} \mathbb{R}^1 \cong \mathbb{R}^2 / \text{im} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cong \mathbb{R}^1$.

Proposition 3.19.

Let V be an $\mathbb{F}$ -vector space. $\mathbb{F}[x]$ acts on V via $T : V \rightarrow V$ and $\mathbb{F}[x]$ acts on $\mathbb{F}$ via 0 . Then

$$V \otimes_{\mathbb{F}[x]} \mathbb{F} \rightarrow V / \text{im } T = \text{coker } T \quad (3.17)$$

and $V \otimes \mathbb{F} \rightarrow V / \text{im } T$ is a balanced isomorphism.

(This was absolutely not clear in class, prove it on your own)

Proposition 3.20.

Let R be commutative $R \otimes_R M \cong M$

Proposition 3.21.

Let A, B are right M modules and C be a right module.

$$(A \oplus B) \otimes C \cong (A \otimes C) \oplus (B \otimes C) \quad (3.18)$$

$$A \otimes (B \oplus C) \cong (A \otimes B) \oplus (A \otimes C) \quad (3.19)$$

Proposition 3.22.

Let R be a commutative I, J ideals, Then

$$R/I \otimes_R R/J \cong R/I + J \quad (3.20)$$

Proof. Homework 5. □

3.2 Algebra Structures

Definition 3.23.

An R -algebra structure on a ring A is a ring homomorphism $\varphi : R \rightarrow A$.

Proposition 3.24.

Let R be a ring and A an R -algebra. Then A is an R - R -bimodule.

Proposition 3.25.

Let R be commutative and A and B be R -algebras. Then $A \otimes_R B$ is a ring with

$$(a \otimes b)(a' \otimes b') = (aa') \otimes (bb') \quad (3.21)$$

Example 3.6

If R is a subring of A , then A is an R -algebra.

Proposition 3.26.

We have

$$R[x] \otimes_R A \cong A[x] \quad (3.22)$$

$$R^n \otimes_R A \cong A^n \quad (3.23)$$

Definition 3.27.

Let R be commutative. Let S be multiplicatively closed subset of R containing 1. Then

$$S^{-1}R = R \times S / \sim \quad (3.24)$$

Where $r/s \sim r'/s'$ if $rs't = r'st$ for some $t \in R$.

Proposition 3.28.

If R is a commutative ring, then

$$M \otimes_R N \cong N \otimes_R M \quad (3.25)$$

We also have associativity of the tensor product with the following proposition:

Proposition 3.29.

$$(A \otimes_R B) \otimes_S C \cong A \otimes_R (B \otimes_S C) \quad (3.26)$$

Definition 3.30: Bilinearity.

Let R be commutative. Then we say $f : M \times N \rightarrow P$ is bilinear if $f(\cdot, n)$ and $f(m, \cdot)$ are both R -linear.

Proposition 3.31.

Let $f : M \times N \rightarrow P$ be R -bilinear. Then there exists a unique $\hat{f} : M \otimes_R N \rightarrow P$ with Figure 4

$$\begin{array}{ccc} M \times N & \xrightarrow{\quad} & P \\ \downarrow \iota & \nearrow \hat{f} & \uparrow f \\ M \otimes N & & \end{array}$$

Figure 4

Proposition 3.32.

Let R, S be rings. Let X, Y, Z be R, S -bimodules. Then

$$\text{hom}_S(Y \otimes_R X, Z) \cong \text{hom}_R(Y, \text{hom}_S(X, Z)) \quad (3.27)$$

4 Linear and Multi-Linear Algebra

Definition 4.1.

Let V be an $\mathbb{F}$ -vector space. A set $S \subseteq V$ is linearly independent if for all $\vec{v}_1, \dots, \vec{v}_n \in S$ and $\alpha_1, \dots, \alpha_n \in \mathbb{F}$, the statement

$$\sum_{i=1}^n \alpha_i \vec{v}_i = \vec{0} \quad (4.1)$$

implies all $\alpha_i = 0$.

Proposition 4.2.

S is a basis of V if

(a) $\langle S \rangle = V$

(b) S is linearly independent.

Definition 4.3.

$S \subseteq V$ is a basis of V if $F_{\mathbb{F}}(S) \cong V$.

Proposition 4.4.

Let $A \subseteq V$ be finite and assume $\langle A \rangle = V$. Then there exists some $B \subseteq A$ with B being a basis for V .

Lemma 4.5.

Let $\{w_i\}_{i=1}^m$ be a basis of $W \subseteq V$. Then there are $v_{m+1}, \dots, v_n$ with $\{w_i\}$ as a basis.

Definition 4.6.

We define the dimension of a vector space V to be $\dim V = n$ if $V \cong \mathbb{F}^n$.

Proposition 4.7.

If $\dim V = \dim W < \infty$ then $V \cong W$.

Proposition 4.8.

Let $W \subseteq V$ be a subspace, then

$$\dim V = \dim W + \dim V/W \quad (4.2)$$

Corollary 4.9: Rank-Nullity Theorem.

Let $\varphi : V \rightarrow U$ be $\mathbb{F}$ -linear. Then

$$\dim V = \dim \ker \varphi + \dim \operatorname{im} \varphi \quad (4.3)$$

Proposition 4.10.

Let $\varphi : V \rightarrow W$. The following are equivalent:

(a) φ is an isomorphism

(b) φ is injective

(c) φ is surjective

(d) φ sends a basis of V to a basis of W .

Definition 4.11: Dual Space.

Let V be an $\mathbb{F}$ -vector space. Let $V^* = \operatorname{hom}(V, \mathbb{F})$.

Proposition 4.12.

If $\dim V < \infty$, then $V \cong V^*$.

Proof. Let $\dim V = n$. Then $V \cong \mathbb{F}^n$ and we can say that

$$\operatorname{hom}_{\mathbb{F}}(V, \mathbb{F}) \cong \operatorname{hom}_{\mathbb{F}}(\mathbb{F}^n, \mathbb{F}) \cong \operatorname{Mat}_{\mathbb{F}}(1, n) \cong \mathbb{F}^n \quad (4.4)$$

by Theorem ??

□

Definition 4.13.

If $\vec{v}_1, \dots, \vec{v}_n$ is a basis of V , let $\vec{v}_i^*$ be

$$\vec{v}_1^* \left(\sum_j a_j \vec{v}_j \right) = a_j \quad (4.5)$$

where we can also say

$$\vec{v}_i^*(\vec{v}_j) = \delta_{ij} \quad (4.6)$$

where δ_{ij} is the Kronecker delta.

Note that there is an isomorphism from $v_i \mapsto v_i^*$ if $\dim V < \infty$.

Proposition 4.14.

If $a_1, \dots, a_n$ is a basis of V , and $b_1, \dots, b_m$ are linearly independent, then $m \leq n$ and $b_1, \dots, b_m, a_{m+1}, \dots, a_n$ is a basis of V after the proper reordering of indices.

Corollary 4.15.

If $W \subseteq V$, then $\dim W \leq \dim V$.

Proof. Any basis $b_1, \dots, b_n$ of W is linearly independent, so apply Proposition 4.14 on a basis $a_1, \dots, a_n$ of V , to see that the basis of V has at least as many elements as the basis of W . $\square$

Note that if $\dim W = \dim V$, then $W = V$.

Definition 4.16.

Let $\alpha = \{\vec{a}_i\}_1^n$ be a basis of V and $\beta = \{\vec{b}_i\}_1^n$ be a basis of W . For $T : V \rightarrow W$ linear, let

$$M_\beta^\alpha(\varphi) = \begin{bmatrix} r_{11} & \cdots & r_{1n} \\ \vdots & \ddots & \vdots \\ r_{m1} & \cdots & r_{mn} \end{bmatrix} \quad (4.7)$$

With

$$T(\vec{a}_j) = \sum_{i=1}^n r_{ij} \vec{b}_i \quad (4.8)$$

Proposition 4.17.

Let $\varphi, \psi : V \rightarrow W$. If $M_\beta^\alpha(\varphi) = M_\beta^\alpha(\psi)$, then $\varphi = \psi$.

Proof. This is a one liner. We see

$$\varphi\left(\sum_i s_i \vec{a}_i\right) = \sum_i s_i \varphi(\vec{a}_i) = \sum_i s_i \psi(\vec{a}_i) = \psi\left(\sum_i s_i \vec{a}_i\right) \quad (4.9)$$

$\square$

Proposition 4.18.

Let M be $m \times n$. Pick bases α on V and β on W . Let $\varphi_\alpha^\beta : V \rightarrow W$ be

$$\varphi_\alpha^\beta\left(\sum_i s_i \vec{a}_i\right) = \sum_i \sum_j s_i r_{ij} \vec{b}_j \quad (4.10)$$

Note that φ_α^β is well-defined and a linear map. Well-definedness follows from Corollary 4.20 below.

Proposition 4.19.

If $\{\vec{a}_i\}_{i=1}^n$ is a basis of V , then for all $\vec{v} \in V$, there exists a unique $\{r_i\}_{i=1}^n \in \mathbb{F}$ with

$$\vec{v} = \sum_{i=1}^n r_i \vec{a}_i \quad (4.11)$$

Corollary 4.20.

Let $\dim U = n$ and $\dim W = M$. Then

$$\text{Mat}_{\mathbb{F}}(n, m) \cong \text{hom}_{\mathbb{F}}(V, W) \quad (4.12)$$

Corollary 4.21.

$$\dim \text{hom}(V, W) = \dim V \dim W \quad (4.13)$$

Definition 4.22.

Let A be an $m \times n$ matrix and let $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ be

$$\varphi_{\{e_1, \dots, e_n\}}^{\{e_1, \dots, e_n\}} \quad (4.14)$$

Example 4.1

If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then we have T_A defined as

$$T_A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ 3x + 4y \end{bmatrix}$$

Definition 4.23.

Let $B = [b_1 \ \dots \ b_n]$ be a matrix whose columns are vectors b_i and A an $m \times n$ matrix. Then define the multiplication by

$$AB = [Ab_1 \ \dots \ Ab_n] \quad (4.15)$$

Proposition 4.24.

$$T_{AB} = T_A \circ T_B \quad (4.16)$$

Proposition 4.25.

Let $V \xrightarrow{\varphi} W \xrightarrow{\psi} U$ with α, β, γ being basis for V, W, U respectively. Then we have

$$M_\alpha^\gamma(\psi \circ \varphi) = M_\beta^\gamma(\varphi) M_\alpha^\beta(\varphi) \quad (4.17)$$

Definition 4.26: Columnspace.

We define the columnspace of A by

$$\text{col}(A) = \text{im } T_A \quad (4.18)$$

Proposition 4.27.

Let $A = [\vec{a}_1 \ \dots \ \vec{a}_n]$. Then

$$\text{col}(A) = \langle a_1, \dots, a_n \rangle \quad (4.19)$$

Definition 4.28: Nullspace.

We define the nullspace of A as

$$\text{Null}(A) = \ker T_A \quad (4.20)$$

The nullity of A is defined as

$$\text{nullity}(A) = \dim \ker T_A \quad (4.21)$$

Proposition 4.29: Rank-Nullity (Formalized).

Let A be $m \times n$. Then $T_A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ and

$$\text{nullity}(A) + \text{rank}(A) = n \quad (4.22)$$

Definition 4.30: Invertibility.

We say that an $n \times n$ matrix A is invertible if there exists some B such that $AB = I_n$.

Proposition 4.31.

Let A be $n \times n$. The following are equivalent:

- (a) A is invertible
- (b) T_A is an isomorphism
- (c) $\text{rank } A = n$
- (d) $\text{nullity } A = 0$

4.1 Adjoint and Determinants

Definition 4.32: Adjoint Map.

Let $T : V \rightarrow W$ and $T^* : W^* \rightarrow V^*$ be $f \in W^*$ with $f : W \rightarrow \mathbb{F}$. Let $T^*(f) \in V^*$ be

$$T^*(f)(\vec{v}) = f(T(\vec{v})) \quad (4.23)$$

Definition 4.33: Symmetric Group.

Define the Symmetric group on n elements by

$$S_n = \{f : \{1, \dots, n\} \rightarrow \{1, \dots, n\} | f \text{ is bijective}\} \quad (4.24)$$

Example 4.2

Define the following:

$$\sigma : 1 \mapsto 2, 2 \mapsto 3, 3 \mapsto 1 \quad (4.25)$$

$$\tau : 1 \mapsto 2, 2 \mapsto 1, 3 \mapsto 3 \quad (4.26)$$

and define

$$\Delta = (x_1 - x_2)(x_1 - x_3)(x_2 - x_3) \quad (4.27)$$

Then

$$\sigma(\Delta) = (x_2 - x_1)(x_2 - x_3)(x_3 - x_1) \quad (4.28)$$

$$= (x_2 - x_3)(-1)(x_1 - x_2)(-1)(x_1 - x_3) \quad (4.29)$$

$$= \Delta \quad (4.30)$$

and

$$\tau(\Delta) = -\Delta \quad (4.31)$$

by the same logic. The goal of this example is to define the sgn function.

Proposition 4.34.

For all $\sigma \in S_n$, we have

$$\sigma(\Delta) = \pm \Delta \quad (4.32)$$

Definition 4.35: Sign Function.

Let $\text{sgn} : S_n \rightarrow \{\pm 1\}$ be

$$\text{sgn}(\sigma) = \frac{\sigma(\Delta)}{\Delta} = \begin{cases} 1 & \text{if } \sigma(\Delta) = \Delta \\ -1 & \text{if } \sigma(\Delta) = -\Delta \end{cases} \quad (4.33)$$

Proposition 4.36.

sgn is a homomorphism.

Proposition 4.37.

$\text{sgn} : S_n \rightarrow \{\pm 1\}$ is surjective for all $n \geq 2$.

Definition 4.38.

Let

$$A_n = \ker(S_n \rightarrow \{\pm 1\}) \quad (4.34)$$

Definition 4.39.

Let $A = (a_{ij}) \in \text{Mat}_{\mathbb{F}}(n, n)$. Then

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{a\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)} \quad (4.35)$$

Theorem 4.40: Determinant.

We have that the determinant defined as

$$\det : \text{Mat}_{\mathbb{F}}(n, n) \rightarrow \mathbb{F} \quad (4.36)$$

is the unique alternating n -linear function sending $\mathbb{I} \mapsto 1$

Note that we often identify $(\mathbb{F}^n)^n \cong \text{Mat}_{\mathbb{F}}(n, n)$.

Definition 4.41.

For $-1 \neq 1 \in \mathbb{F}$, we say

$$f : V \times \cdots \times V \rightarrow W \quad (4.37)$$

is alternating if $f(\vec{v}_1, \dots, \vec{v}_n) = \text{sgn}(\sigma)f(\vec{v}_{\sigma(1)}, \dots, \vec{v}_{\sigma(n)})$ for all $\sigma \in S_n$.

Example 4.3

The usual 3-dimensional cross product defined as

$$\times : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3 \quad (\vec{v}, \vec{w}) \mapsto \vec{v} \times \vec{w}$$

is alternating.

Proposition 4.42.

The determinant as defined in Theorem 4.40 is alternating.

Proposition 4.43.

The determinant is n -linear and $\det \mathbb{I}_n = 1$ for all n .

Proposition 4.44.

If $f : \mathbb{F}^{m_1} \times \cdots \times \mathbb{F}^{m_n} \rightarrow W$ is n -linear, then f is determined by its values on $(e_{i_1}, \dots, e_{i_n})$.

Proof. See Example 4.1. □

Example 4.4

Define $f : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ and let

$$f(e_1, e_1) = 2$$

$$f(e_1, e_2) = 3$$

$$f(e_2, e_1) = 4$$

$$f(e_2, e_2) = 0$$

What is $f\left(\begin{bmatrix} c \\ d \end{bmatrix}, \begin{bmatrix} a \\ b \end{bmatrix}\right)$? We see

$$\begin{aligned} f\left(\begin{bmatrix} c \\ d \end{bmatrix}, \begin{bmatrix} a \\ b \end{bmatrix}\right) &= f(ae_1 + be_2, ce_1 + de_2) \\ &= af(e_1, ce_1 + de_2) + baf(e_2, ce_1 + de_2) \\ &= acf(e_1, e_1) + adf(e_1, e_2) + bcf(e_2, e_1) + bdf(e_2, e_2) \\ &= 2ac + 3ad + 4bc \end{aligned}$$

The proof of Proposition 4.44 is given by the same computation in more general terms.

Proposition 4.45.

$\det(A) = \det(A^T)$.

Theorem 4.46: Expansion by Minors.

We can expand the determinant in the following way:

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}) \quad (4.38)$$

$$= \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}) \quad (4.39)$$

Theorem 4.47.

Let R be commutative, and f bilinear. Then there exists a unique $\hat{f}$ which is linear and makes the diagram in Figure 5 commute.

$$\begin{array}{ccc} M_1 \times M_2 & \xrightarrow{f} & W \\ \downarrow & \searrow \hat{f} & \\ M_1 \otimes_R M_2 & & \end{array}$$

Figure 5

This theorem generalizes to k elements.

Definition 4.48.

Define

$$\text{sym}^k(M) = (M \otimes_R \cdots \otimes_R M) / N \quad (4.40)$$

where

$$N = \langle (m_1 \otimes \cdots \otimes m_k) - (m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)}) \rangle \quad (4.41)$$

Then we have

$$\text{Sym}(M) = \bigoplus_k \text{Sym}^k(M) \quad (4.42)$$

Definition 4.49.

We say that $f : M^k \rightarrow W$ is symmetric if

$$f(m_1, \dots, m_k) = f(m_{\sigma(1)}, \dots, m_{\sigma(k)}) \quad (4.43)$$

for all $\sigma \in S_k$.

Theorem 4.50.

Let R be commutative, and $f : M^k \rightarrow W$ be k -linear and symmetric. Then there exists a unique $\hat{f}$ which is linear and makes the diagram in Figure 6 commute.

$$\begin{array}{ccc} M^k & \xrightarrow{f} & W \\ \downarrow & \searrow \hat{f} & \\ \text{Sym}^k M & & \end{array}$$

Figure 6

4.2 An Excursion into Multivariate Calculus**Definition 4.51.**

Assume $1 \neq -1 \in R$. Then

$$\wedge^k M = M^{\otimes k} / \langle m_1 \otimes \cdots \otimes m_k + m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(k)} \rangle \quad (4.44)$$

Example 4.5

$$a \wedge b = -(b \wedge a)$$

Theorem 4.52.

We define

$$\dim_{\mathbb{F}} \wedge^k \mathbb{F}^n = \binom{n}{k} = \frac{n!}{k!(n-k)!} \quad (4.45)$$

and $S = \{e_{n_1} \wedge \cdots \wedge e_{n_k}\}$ is a basis for $n_i < n_{i+1}$.

Definition 4.53: Cross Product.

We can define

$$\times : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow (\mathbb{R}^3)^{\otimes 2} \rightarrow \wedge^2 \mathbb{R}^3 \cong \mathbb{R}^3 \quad (4.46)$$

as our cross product in 3 dimensions.

Definition 4.54.

Let $f : \mathbb{R}^n \rightarrow \wedge^k \mathbb{R}^n$. Then we define $df : \mathbb{R}^n \rightarrow \wedge^{k+1} \mathbb{R}^n$ as

$$g(e_{n_1} \wedge \cdots \wedge e_{n_k}) \mapsto \sum_i \frac{\partial g}{\partial x_i} e_i \wedge e_{n_1} \wedge \cdots \wedge e_{n_k} \quad (4.47)$$

for some $g : \mathbb{R}^n \rightarrow \mathbb{R}$.

The upshot of this definition is that in $\mathbb{R}^3$ the function $f : \mathbb{R}^3 \rightarrow \wedge^1 \mathbb{R}^3$ is a vector field and df is the curl of f . If instead $f : \mathbb{R}^3 \rightarrow \wedge^2 \mathbb{R}^3$, df becomes the divergence.

5 Matrix Algebra

Definition 5.1.

Let $T : V \rightarrow W$ be linear and $\widehat{T} : \wedge^k V \rightarrow \wedge^k W$ be

$$\widehat{T}(v_1 \wedge \cdots \wedge v_k) = T(v_1) \wedge \cdots \wedge T(v_k) \quad (5.1)$$

Proposition 5.2.

$\widehat{T}$ is linear and well defined. Furthermore, we have that if $T : \mathbb{F}^n \rightarrow \mathbb{F}^n$ has matrix A . Then

$$\widehat{T}(e_1 \wedge \cdots \wedge e_n) = \det(A) e_1 \wedge \cdots \wedge e_n \quad (5.2)$$

Definition 5.3.

Let $T : V \rightarrow V$ where $\dim V = n$. Then $\det(T) = a \in \mathbb{F}$ if $\widehat{T} : \wedge^n V \rightarrow \wedge^n V$ is defined by multiplication by a .

Proposition 5.4.

Let $T : V \rightarrow W$ and $S : W \rightarrow U$. Then

$$\widehat{S} \circ \widehat{T} = \widehat{(S \circ T)} : \wedge^k V \rightarrow \wedge^k U \quad (5.3)$$

Corollary 5.5.

If $S, T : V \rightarrow V$, then

$$\det(S \circ T) = \det(S) \det(T) \quad (5.4)$$

Corollary 5.6.

$$\det(T^{-1}) = \frac{1}{\det(T)} \quad (5.5)$$

Proof.

$$1 = \det(\mathbb{I}) = \det(TT^{-1}) = \det(T) \det(T^{-1})$$

□

Corollary 5.7.

If $\det(T) = 0$, then T is not invertible.

Proposition 5.8.

If $T : V \rightarrow V$ is not invertible, then $\det T = 0$.

Theorem 5.9.

Let R be a PID. If M is finitely generated, then

$$M \cong F/(r_1) \oplus \cdots \oplus R/(r_k) \quad (5.6)$$

Definition 5.10.

Let A be a $\mathbb{Z}$ module. A is divisible if for all $a \in A$ and all $n \in \mathbb{N}$, there is a $b \in A$ such that $nb = a$.

Proposition 5.11.

Let $A \xrightarrow{\varphi} B \xrightarrow{T} C \xrightarrow{\psi} D$ be linear, with φ and ψ isomorphisms. Then

$$\text{coker}(T) \cong \text{coker}(\psi \circ T \circ \varphi) \quad (5.7)$$

Corollary 5.12.

If A is obtained from B via row and column operations, then $\text{coker } A \cong \text{coker } B$.

Theorem 5.13.

View $M = \mathbb{F}^n$ as an $\mathbb{F}[x]$ module with x acting via $A \in \text{Mat}_{\mathbb{F}}(n, n)$. Then $M \cong \text{coker}(A - xI)$ as $\mathbb{F}[x]$ -modules.

Definition 5.14: Similar Matrices.

We say that A is similar to C , $A \sim C$, if there exists some B with $A = BCB^{-1}$

Theorem 5.15.

$A \sim C$ if and only if $\mathbb{F}^n$ with x acting via A is isomorphic to $\mathbb{F}^n$ acting with C .

Theorem 5.16: Structure Theorem.

Let M be a finitely generated module over a PID. Then

$$M = \bigoplus_{i=1}^k R/(r_i) \quad (5.8)$$

5.1 Noetherian Rings and Modules

Definition 5.17: Noetherian Ring.

We say that R is Noetherian if for every $I_1 \subseteq I_2 \subseteq \cdots$, then there is some k with $I_n = I_k$ for all $n \geq k$.

Proposition 5.18.

If R is a PID, then R is Noetherian.

Proof. Let $I = \bigcup_{j=1}^{\infty} I_j$, and $I = (a)$ for all $a \in I$. SO then there must be some k with $a \in I_k$. Since $a \in I_n$ so then $(a) \subseteq I_k \subseteq I_n \subseteq I = (a)$ and we are done. □

Definition 5.19: Noetherian Module.

Let M be a left R -module. M is Noetherian if for all $M_1 \subseteq M_2 \subseteq \cdots$ there exists a k with $M_n = M_k$ for all $n \geq k$,

Theorem 5.20.

M is Noetherian if and only if all submodules of M are finitely generated

Proof. ($\Rightarrow$) Pick some $N \subseteq M$ that is not finitely generated, and $n_0 \in N$. $\langle n \rangle \neq N$ since it is not finitely generated, so for some $n_1 \in N$, we have $n_1 \in \langle n_0 \rangle$. Again, since N is not finitely generated, we know $\langle n_0, n_1 \rangle \neq N$ so then there is some $n_2 \in N$ such that $n_2 \in \langle n_0, n_1 \rangle$. We can continue this process forever, so M cannot be Noetherian. Therefore, if all submodules are finitely generated, then the M is Noetherian.

($\Leftarrow$) Assume M is not Noetherian. Pick some $M_0 \subsetneq M_1 \subsetneq M_2 \subsetneq \cdots$. Pick some $m_i \in M_i$ such that $m_i \notin M_{i-1}$. Let $N = \langle m_0, m_1, \dots \rangle$. If $N = \langle n_0, n_1, \dots, n_k \rangle$ then each $n_i \in M_{n_i}$ for some i . Let $J = \max_i \{n_i\}$. Then $n_i \in M_J$ for all $i = 0, \dots, k$ so $N \subseteq M_J$. But we also have $m_{J+1} \in M \subseteq M_J$. But by our assumptions, $M_{J-1} \not\subseteq M_J$ so then N cannot be finitely generated. □

Proposition 5.21.

If A and B are Noetherian, so is $A \oplus B$.

Proof. Pick some $M_1 \subseteq M_2 \subseteq \cdots \subseteq A \oplus B$. Let

$$A_i = (M_i + B)/B \subseteq A \cong (A \oplus B)/B \quad (5.9)$$

$$B_i = (M_i + A)/A \subseteq B \cong (A \oplus B)/A \quad (5.10)$$

Then there exists some k, ℓ such that $A_n = A_k$ for $n \geq k$ and $B_m = B_\ell$ for all $m \geq \ell$. Let $P = \max\{k, \ell\}$. Pick $m \in M_j$ for $j \geq p$ and $m = a + b$ for $a \in A_p$ and $b \in B_p$. Then there exists some $\hat{a} \in A$ and $\hat{b} \in B$ such that $a + \hat{b} \in M_p$ and $b + \hat{a} \in M_p$. The direct sum is unique, so then $\hat{b} = b$ and $\hat{a} = a$, so then $a + b \in M_p$ and $M_j \in M_p$ so $A \oplus B$ is indeed Noetherian. $\square$

Corollary 5.22.

If R is Noetherian, R^n is Noetherian.

Proposition 5.23.

If M is Noetherian, $f : M \rightarrow N$, then N is Noetherian. Let $N' \subseteq N$. $f^{-1}(N') \subseteq M$ is finitely generated. Let $\langle m_1, \dots, m_k \rangle = f^{-1}(N')$. Then

$$N' = \langle f(m_1), \dots, f(m_k) \rangle \quad (5.11)$$

Proposition 5.24.

Let R be Noetherian and M be finitely generated. Then M is Noetherian.

Corollary 5.25.

Let R be Noetherian, M finitely generated and $N \subseteq M$. Then N is finitely generated.

Definition 5.26: Finitely Presented Module.

M is finitely presented if

$$M \cong \text{coker } (g : R^m \rightarrow R^n) \quad (5.12)$$

Proposition 5.27.

If R is Noetherian and M is finitely generated, then M is finitely presented.

Theorem 5.28.

If R is Noetherian and M finitely generated, then

$$M = \text{coker } (A : R^n \rightarrow R^m) \quad (5.13)$$

5.2 Diagonalizability and the Smith Algorithm

Theorem 5.29.

If R is Euclidean, then A is row and column equivalent to a diagonal matrix.

Theorem 5.30.

If R is a PID and A an $m \times n$ matrix, then there exists invertible X and Y such that XAY is diagonal.

Corollary 5.31.

If R is a PID and M finitely generated, then

$$M \cong \bigoplus_{i=1}^n R/(r_i) \quad (5.14)$$

Theorem 5.32: Invariant Factor Decomposition.

If R is a PID and M a finitely generated module then

(a) $M \cong R^n \oplus R/(a_1) \oplus \cdots \oplus R/(a_m)$ such that $a_1 | a_2 | \cdots | a_m$. This is called the invariant factor decomposition of M .

(b) $M \cong R \oplus R/p_1^{n_1} \oplus \cdots \oplus R/p_\ell^{n_\ell}$ where p_i are primes. This is called the primary decomposition of M .

Corollary 5.33.

If $(a, b) = 1$, then $R/a \oplus R/b = R/ab$.

Definition 5.34: Torsion Module.

We define the torsion module of M by

$$\text{Tor}(M) = \{m \in M : \exists r \neq 0, rm = 0\} \quad (5.15)$$

Example 5.1

$$\text{Tor}(\mathbb{Z}/2 \oplus \mathbb{Z}) = \mathbb{Z}/2$$

Proposition 5.35.

$\text{Tor}(M)$ is a submodule of M .

Proposition 5.36.

If M is a finitely generated module over a PID R , then

$$M/\text{Tor}(M) \cong R^n \quad (5.16)$$

Proposition 5.37.

If R is an integral domain and $R^n \cong R^m$, then $n = m$.

Proof. We know that $S = R \setminus \{0\}$, then $F = S^{-1}R$ and

$$R^n \cong R^m \implies F^n = S^{-1}R^n \cong S^{-1}R^m = F^m \quad (5.17)$$

so then $n = m$. □

Proposition 5.38.

Let R be an integral domain. If $M \cong R^n \oplus A \cong R^m \oplus B$ and A, B are torsion modules, then $A \cong B$ and $n = m$.

Proposition 5.39.

Let R be a PID, and M finitely generated. The primary decomposition and elementary factor decomposition are unique.

Theorem 5.40.

Let $\mathbb{F}^n \cong_{\mathbb{F}[x]} \text{coker}(\mathbb{F}[x]^n \rightarrow \mathbb{F}[x]^n)$. Then $A - xI$ is row or column wise diagonalizable, such that $a_1(x)|a_2(x)|\dots|a_n(x)$ and

$$\mathbb{F}^n \cong \bigoplus_{i=1}^n \mathbb{F}[x]^n / a_i(x) \quad (5.18)$$

Theorem 5.41.

Let $\mathbb{F}[x]$ act on $\mathbb{F}^n$ via A . Then $M \cong \bigoplus_{i=1}^n \mathbb{F}[x]/x - \lambda_i$ if and only if A is diagonalizable with eigenvalues λ_i .

Theorem 5.42.

$$\mathbb{F}^n \cong \text{coker}(A - xI : \mathbb{F}^n[x] \rightarrow \mathbb{F}^n[x]) \quad (5.19)$$

Definition 5.43: Characteristic Polynomial.

We define the characteristic polynomial of A as

$$P_A(x) = \pm \det(A - xI) \quad (5.20)$$

Definition 5.44: Minimal Polynomial.

The minimal polynomial of A , $m_A(x)$ is the monic generator of the set

$$\{f : f(A) = 0\} \quad (5.21)$$

Theorem 5.45.

For any $n \times n$ matrix A , we have $m_A(x) | P_A(x)$

Proof. This is the same as showing $P_A(A) = 0$. We know from Theorem 5.42 that

$$\mathbb{F}^n \cong \text{coker } (A - xI : \mathbb{F}^n[x] \rightarrow \mathbb{F}^n[x]) \quad (5.22)$$

which implies that

$$A - x \sim \begin{bmatrix} P_1(x) & & \\ & \ddots & \\ & & P_n(x) \end{bmatrix} \quad (5.23)$$

From here, we have

$$P_A(x) = \det(A - x) = \prod_{i=1}^n P_i(x) \quad (5.24)$$

We then know

$$\mathbb{F}^n \cong \bigoplus_{i=1}^n \mathbb{F}[x]/(P_i(x)) \quad (5.25)$$

If we pick some $\vec{v} \in \mathbb{F}[x]/P_j(x)$, we can write $\vec{v} = \sum_j r_j \vec{v}_j$ for some basis $\{\vec{v}_j\} \in \mathbb{F}[x]/(P_j(x))$ and $\{r_j\} \in \mathbb{F}$, then $\left(\prod_{j=1}^n P_j(x)\right) \vec{v}_j = \vec{0}$, which means $P_A(A) = 0$ and $P_A(x)$ is a multiple of the minimal polynomial. $\square$

Definition 5.46: Diagonalizability.

An $n \times n$ matrix A is diagonalizable if it is similar to a diagonal matrix.

Definition 5.47: Eigenvalues and Eigenvectors.

We say that λ is an eigenvalue of A if there exists some $\vec{v} \neq \vec{0}$ such that $A\vec{v} = \lambda\vec{v}$. Such a $\vec{v}$ is called λ 's associated eigenvector.

Definition 5.48: Eigenspace.

We say that the eigenspace of A for an eigenvalue λ is given by

$$E_\lambda = \ker(A - \lambda I) \quad (5.26)$$

We call $\dim E_\lambda$ the geometric multiplicity of λ .

Definition 5.49: Algebraic Multiplicity.

If $P_A(x) = \prod_i (x - \lambda_i)^{n_i}$, then we say that $P_A(x)$ splits.

Definition 5.50: Algebraic Multiplicity.

If $P_A(x)$ splits, and all λ_i are distinct, then the algebraic multiplicity of λ_i is n_i .

Proposition 5.51.

Suppose $P_A(x)$ splits. Then A is diagonalizable if and only if the geometric and algebraic multiplicities are equal for all eigenvalues.

Proposition 5.52.

If λ is an eigenvalue, then

$$1 \leq \text{geom } \lambda \leq \text{alg}(\lambda) \quad (5.27)$$

Proposition 5.53.

Suppose $\mathbb{F}^n \cong \bigoplus_i \mathbb{F}[x]/(x - \lambda_i)^{n_i}$. Then

$$m_A(x) = \prod_j (x - \lambda_j)^{m_j} \quad (5.28)$$

where λ_j are distinct, and $m_j = \max\{n_i : \lambda_i = \lambda_j\}$.

Corollary 5.54.

If $P_A(x)$ splits and $P_A(\lambda) = 0$, then $m_A(\lambda) = 0$.

6 Canonical Matrix Forms

Proposition 6.1.

Let $V = \mathbb{F}[x]/p(x)$ where $p(x)$ is a monic polynomial of degree d with coefficients r_i . Let $T : V \rightarrow V$ be multiplication by x . Then $\{1, x, x^2, \dots, x^{d-1}\}$ is a basis of V and the matrix of T with respect to this basis is

$$C_{p(x)} = \begin{bmatrix} 0 & 0 & 0 & \dots & -r_0 \\ 1 & 0 & 0 & \dots & -r_1 \\ 0 & 1 & 0 & \dots & -r_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & -r_{d-1} \end{bmatrix} \quad (6.1)$$

Corollary 6.2: Rational Canonical Form.

Let A be an $n \times n$ matrix. Then there exists unique $a_1(x), \dots, a_k(x)$ with $a_1(x) | a_2(x) | \dots | a_k(x)$ with

$$A \sim \begin{bmatrix} C_{a_1(x)} & & \\ & \ddots & \\ & & C_{a_k(x)} \end{bmatrix} \quad (6.2)$$

Definition 6.3: Jordan Block.

We define the Jordan block of size d , by

$$J_d(\lambda) = \begin{bmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \lambda & 1 & \\ & & & \lambda & 1 \\ & & & & \ddots & \ddots \\ & & & & & \lambda & 1 \end{bmatrix} \quad (6.3)$$

Theorem 6.4.

If $P_A(x)$ splits, then

$$A \sim \begin{bmatrix} J_{d_1}(\lambda_1) & & \\ & J_{d_2}(\lambda_2) & \\ & & \ddots \end{bmatrix} \quad (6.4)$$

which is Jordan Normal form.

Proposition 6.5.

For A_i square, we have

$$\begin{bmatrix} A_1 & & \\ & A_2 & \\ & & \ddots \end{bmatrix}^k = \begin{bmatrix} A_1^k & & \\ & A_2^k & \\ & & \ddots \end{bmatrix} \quad (6.5)$$

Using this proposition, we can see how we can raise any matrix to a high power, and easily calculate it using Jordan normal form.

Proposition 6.6.

$$[J_d(\lambda)]^k = \begin{bmatrix} \lambda^k & k\lambda^{k-1} & \binom{k}{2}\lambda^{k-2} & \dots & \binom{k}{d}\lambda^{k-d} \\ & \lambda^k & k\lambda^{k-1} & \dots & \binom{k}{d-1}\lambda^{k-(d-1)} \\ & & \lambda^k & \dots & \binom{k}{d-2}\lambda^{k-(d-2)} \\ & & & \ddots & \vdots \\ & & & & \lambda^k \end{bmatrix} \quad (6.6)$$

Definition 6.7.

We say that λ is an eigenvalue of A if $A\vec{v} = \lambda\vec{v}$ for $\vec{v} \neq \vec{0}$. We say that $\vec{v}$ is its associated eigenvector.

Definition 6.8.

The space of eigenvectors for an eigenvalue λ is given by

$$E_\lambda = \ker(A - \lambda I) \quad (6.7)$$

For consistency, we say that $E_\lambda \setminus \{\vec{0}\}$ are all of the (nontrivial) λ eigenvectors.

We can now use Jordan normal form to determine the number of eigenvalues of a matrix with the following proposition:

Proposition 6.9.

If $P_A(x)$ splits, then $\dim E_\lambda$ is equal to the number of λ Jordan blocks.

Definition 6.10: Generalized Eigenvector.

Let $\vec{v}_1$ be a λ -eigenvector. For $j \geq 2$, solve $A\vec{v}_j = \lambda\vec{v}_j + \vec{v}_{j-1}$, we say that $\vec{v}_j$ is a depth j generalized eigenvector associated to $\vec{v}_1$.

Theorem 6.11.

Assume $P_A(x)$ splits. Let $B = \{b_1, \dots, b_n\}$ be a maximal linearly independent set of generalized eigenvectors. Then $A = BJB^{-1}$ with J in Jordan Normal Form and B is some $B = [b_1, \dots, b_n]$. Note that the b_i may have to be normal to determine J in JNF.

Theorem 6.12.

$$\text{nullity}(A - \lambda I)^k = \#_J^k \quad (6.8)$$

Where $\#_J^k$ is the number of Jordan blocks of size k or greater.

Theorem 6.13.

A is diagonalizable if and only if $\sum_\lambda \dim E_\lambda = n$ of and only if $\mathbb{F}^n \cong \bigoplus_i \mathbb{F}[x]/(x - \lambda_i)$.

Lemma 6.14.

Let $\mathbb{F}$ be a field, and $0 \neq f(x) \in \mathbb{F}[x]$. Then there exists some field $\mathbb{K} \supset \mathbb{F}$ with $f(x) = g(x)(x - \lambda) \in \mathbb{K}[x]$.

Corollary 6.15.

Let $\mathbb{F}$ be a field, $f(x) \in \mathbb{F}[x]$. Then there exists some field $K \supset \mathbb{F}$ such that $f(x)$ splits over $\mathbb{K}$.

7 Spectral Theory

Definition 7.1.

Let $f_A : \mathbb{F}^n \times \mathbb{F}^n \rightarrow \mathbb{F}$ is

$$f_A(\vec{v}, \vec{w}) = \vec{v} \cdot (A\vec{w}) \quad (7.1)$$

We say f is alternating if $f(\vec{v}, \vec{w}) = -f(\vec{w}, \vec{v})$ and symmetric if $f(\vec{v}, \vec{w}) = f(\vec{w}, \vec{v})$.

Example 7.1

Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then

$$f_A \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, [1, 0] \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = a$$

If f_A is alternating, then $a = 0$ and $d = 0$, so $A = -A^T$. If f_A is symmetric, then $A = A^T$.

This example gives us equivalent definitions to symmetric and alternating matrices in the form of the following proposition:

Proposition 7.2.

Assume we are not in a field of characteristic 2. Then f_A is symmetric if and only if A is symmetric and f_A is alternating if and only if A is anti-symmetric.

Definition 7.3: Inner Product.

Let V be an $\mathbb{R}$ -vector space. Then $f : V \times V \rightarrow \mathbb{R}$ is an inner product if:

- (a) f is bilinear.
- (b) f is symmetric.
- (c) $f(\vec{v}, \vec{v}) \geq 0$ for all $\vec{v} \neq 0$.

Example 7.2

Take $A = \begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix}$. The natural question arises: is f_A an inner product? The eigenvalues of A are 5 and 3, and the corresponding eigenvector to 3 is negative, so it cannot be an inner product.

Theorem 7.4: Spectral Theorem.

Let $A \in \text{Mat}_{\mathbb{R}}(n, n)$ be symmetric. Then there exists a $B \in \text{Mat}_{\mathbb{R}}(n, n)$ with

- (a) BAB^{-1} diagonal.
- (b) $B^T = B^{-1}$.

Corollary 7.5.

If A is symmetric and real, then A is diagonalizable with real eigenvalues.

Corollary 7.6.

f_A is an inner product on $\mathbb{R}^n$ if and only if A is symmetric all eigenvalues are positive.

Proposition 7.7.

Let $f : V \times V \rightarrow \mathbb{R}$ be an inner product. Then f is non-degenerate.

Definition 7.8.

$f : V \times V \rightarrow \mathbb{F}$ is bilinear. Let $f^* : V \rightarrow V^*$ be

$$v \mapsto (w \mapsto f(\vec{v}, \vec{w})) \quad (7.2)$$

Then f is non-degenerate if f^* is an isometry.

Definition 7.9: Isometry.

Let (V, f) and (W, g) be inner product spaces. Then $T : V \rightarrow W$ is an isometry if

- (a) T is an isomorphism.
- (b) $g(T(\vec{v}), T(\vec{w})) = f(\vec{v}, \vec{w})$.

For the sake of notation going forward, let $f : V \times V \rightarrow \mathbb{R}$ be a fixed inner product on V an $\mathbb{R}$ -vector space.

Definition 7.10.

For any subset $W \subseteq V$, define

$$W^\perp = \{\vec{u} \in V : |f(\vec{u}, \vec{w})| = 0 \forall \vec{w} \in W\} \quad (7.3)$$

Proposition 7.11.

We can decompose any V such that

$$V = W \oplus W^\perp \quad (7.4)$$

Corollary 7.12.

For all $\vec{v}$, there exists a unique $\vec{w} \in W$ and $\vec{u} \in W^\perp$ such that $\vec{v} = \vec{w} + \vec{u}$.

Definition 7.13.

Define the projection of $\vec{v}$ onto W by

$$\text{proj}_W \vec{v} = \vec{w} \quad (7.5)$$

If $\vec{v} = \vec{w} + \vec{u}$ for $\vec{w} \in W$ and $\vec{u} \in W^\perp$.

Proposition 7.14.

$\text{proj}_W : V \rightarrow V$ is a linear transformation and

$$(a) \text{ proj}_W \circ \text{proj}_W = \text{proj}_W$$

$$(b) \text{ im}(\text{proj}_W) = W$$

$$(c) \text{ ker}(\text{proj}_W) = W^\perp$$

Definition 7.15.

We say that $\{\vec{v}_j\}_{j=1}^n$ is an orthonormal basis of V if it is a basis of V and

$$f(\vec{v}_i, \vec{v}_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad (7.6)$$

Proposition 7.16.

If

$$f(\vec{v}_i, \vec{v}_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad (7.7)$$

then $\{\vec{v}_j\}_{j=1}^n$ are linearly independent.

Proof. Assume $\vec{0} = \sum_{j=1}^n r_j \vec{v}_j$. We want to show that $r_j \equiv 0$ for all j . □

Proposition 7.17.

If $W \subseteq V$, and $\{\vec{v}_i\}_1^m$ is an orthonormal basis of W , then

$$\text{proj}_W \vec{x} = \sum_{i=1}^m f(\vec{x}, \vec{v}_i) \vec{v}_i \quad (7.8)$$

In my physics courses people call this "Fourier's Trick" which I'm not sure is accurately named, but is the same idea when $m = \infty$ which is what happens when we make a basis for L^1 .

Definition 7.18.

Let (V, f) and (U, g) be inner product spaces and $T : V \rightarrow U$ is a linear operator. Let $T^* : U \rightarrow V$ be defined by Figure 7.

$$U \xrightarrow{g^*} U^* \longrightarrow V^* \xrightarrow{(f^*)^{-1}} V$$

Figure 7

Part II

Previous Qualifying Exams

These are problems from past Purdue qualifying exams which can be found <https://www.math.purdue.edu/academic/grad/qual exams.html>. Note that all solutions are either mine or sourced from other resources, so the validity of them should not be assumed. I have tried to reference relevant theorems in the notes section above, but some results may be assumed as true when they were proved in class or homework.

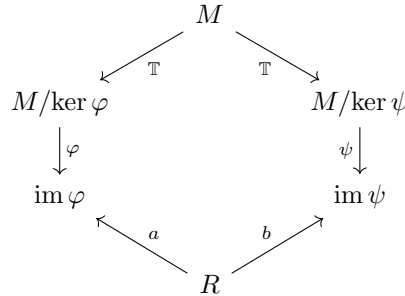
Exam 1: August 2025

Problem 1.1

Let R be a PID, M an R -module, and φ, ψ nonzero R -linear maps $M \rightarrow R$. Prove that $\ker \varphi = \ker \psi$ if and only if $a\varphi = b\psi$ for some nonzero $a, b \in R$.

Solution to Problem 1.1: Assume $a\varphi = b\psi$ and $a, b \neq 0$. Let $x \in \ker \varphi$. We want to show that $x \in \ker \psi$. We know that $\varphi(x) = 0$, so $a\varphi(x) = 0$ and $b\psi(x) = 0 \in R$. But since R is an integral domain, $b \neq 0$ and $\psi(x) = 0$ the kernels coincide.

For the reverse direction, assume $\varphi, \psi \neq 0$ and $\ker \varphi = \ker \psi$. We claim the following diagram helps in some way:



We claim the following:

- (a) If $T : R \rightarrow R$ is R -linear, then there exists γ with $T(r) = \gamma r$ for all r .
- (b) T is an isomorphism if and only if γ is a unit.
- (c) If M, N are isomorphic to R , and $T, S : M \rightarrow N$ is an isomorphisms, then there exists μ a unit with $T\mu S$.

Idk figure it out I guess?

□

Problem 1.2

Let $R = K[T]$ be the polynomial ring in one variable over a field K , let A be an $n \times n$ matrix with entries in R , and let $\varphi : R^n \rightarrow R^n$ be the R -linear map with $\varphi(x) := A \cdot x$, where $x \in R^n$ is a column vector. Write $M = \text{coker } \varphi := R^n / \text{im } \varphi$, which is an R -module and, in particular, a K -vector space via the inclusion $K \subset R$. Prove that:

- (a) $\det A \neq 0$ if and only if M is a torsion R -module.
- (b) if $\det A \neq 0$, then $\dim_K M = \deg(\det A)$, the degree of the polynomial $\det A \in R$

Solution to Problem 1.2:

(a) Let

$$C = \begin{bmatrix} f_1(t) & & & \\ & f_2(t) & & \\ & & \ddots & \\ & & & f_n(t) \end{bmatrix} \quad (1.2.1)$$

where $f_1(t)|f_2(t)\dots f_n(t)$ and A is equivalent to C via row or column operations. Then we know that $\text{coker}(A) \cong \text{coker}(C)$. Since the row and column operations can only change the determinant up to a unit, let $\mu \in K^*$ be a unit and we have

$$\det A = \mu \det C = \mu \prod_{i=1}^n f_i(t) \quad (1.2.2)$$

We also have $\text{coker } C \cong \bigoplus_{i=1}^n R/(f_i(x))$ by the structure theorem. We need to determine when $K[x]/f(x)$ is a torsion module. We claim $f(x) \neq 0 \in K[x]$ if and only if $K[x]/f(x)$ is torsion. Prove this and the result follows.

(b)

□

References

- [1] David S. Dummit and Richard M. Foote. *Abstract Algebra*. 3rd ed. New York: Wiley, 2004.
- [2] Kenneth M. Hoffman and Ray Kunze. *Linear Algebra*. 2nd ed. Englewood Cliffs, NJ: Prentice-Hall, 1971.
- [3] Jeremy Miller. *Linear Algebra Lecture Notes*. URL: <https://drive.google.com/drive/folders/1BTSSPrPdoWek9Bra7z0XIq>